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Biology
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Class 10 [Phase I]

1) (4) Both (1) and (2)

2) (2) Mitochondria only

3) (2) WBC

4) (2) Transpiration

5) (4) 1 and 2

6) (4) None of these

7) Respiration in
PlantsRespiration in
Animals→ occurs only
at night→ occurs at any
time→ They respire
through stomata→ They have different
organs in their body→ Chlorophyll is
required→ Chlorophyll is not
required

8)

Light Reaction

- ~~splitting~~ conversion of light energy into chemical energy and splitting of water into Hydrogen and Oxygen
- No sugar is formed
- O_2 is released

Dark Reaction

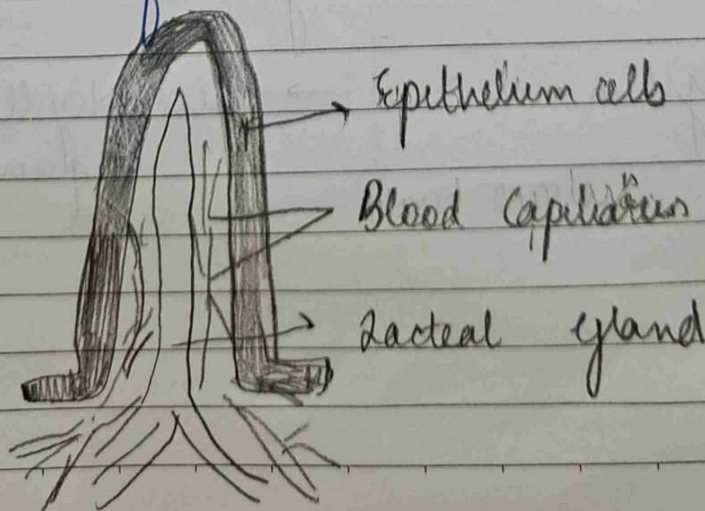
- Reduction of CO_2 into carbohydrates
- Sugar is formed
- No O_2 is released

9 (a) Pepsin, Trypsin ~~are in~~ and ~~are~~ are enzymes that act on proteins

(b) Pyruvate ~~lack~~ ~~absence of~~ $O_2 \rightarrow$ Ethanol + CO_2 + 2ATP

10 (a) Acid in the stomach kills all the harmful germs and bacteria
It creates an acidic medium for pepsinogen to function and convert into pepsin

(b) Lacteals ~~part of villi~~ absorb lipids and fats large molecules from small intestine



11 (a) The alveoli of lungs are covered with blood capillaries because so as to absorb that exchange of gases can take place smoothly. It also produces a large absorptive surface area so that O_2 can be absorbed and transported to various cells in the body.

(b) The walls of trachea is supported by rings of cartilage ^{so as to} prevent it from collapsing.

12 (a) During winter, we sweat less [and sweating is also a way to remove wastes from the body] so due to less of sweating the excess fluid is excreted in the form of urea. Thus, we have more urine produced in winter.

(b) ~~Kid~~ Skin, lungs and liver are accessory excretory organs

(c) ~~1) Nephridia~~ 1) Malpighian Tubes
~~2) Coxal glands~~ 2) Coxal glands
 3) Malpighian tubes 3) Nephridia
 are excretory organs in Arthropoda

3) (i) 4 → Left Atrium
 5 → Right Atrium

(ii) 3 and 6 are superior vena cava and inferior vena cava respectively. Superior vena cava collects all the deoxygenated blood from the top of the body (head, etc) Inferior vena cava on the other hand collect all the deoxygenated blood from lower part of body (legs, etc)

(iii) Arteries are blood vessels which carry blood away from heart

(iv) We can say that circulation in human beings is a double circulation because blood passes the heart twice.

Once when deoxygenated blood enters the R^o right atrium and from right ventricle it is sent to lungs through Pulmonary Artery.

Whereas 2nd one that time when lungs through Pulmonary Vein send oxygenated Blood back to left atrium.

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